Yingrui Hou

Simulation & Model Validation | Digital Twin Applications

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https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience connecting measurements to simulation, calibrating models and diagnosing discrepancies. Built Python/C++/Geant4 workflows; learned TensorFlow through code adaptation and simulation-based verification, then migrated computation to a shared GPU workstation. Brings research coordination and technical communication to simulation engineering and digital twin opportunities.

Core skills

- Simulation: Geant4/C++, detector geometry, interaction stepping, response logic, event-level outputs and measured response maps.
- Validation: simulation-to-measurement alignment, deep-learning calibration, image-recognition model checks, bias correction and uncertainty analysis.
- Signals & models: time-profile decomposition, likelihood inference, CatBoost with five-fold checks, control-sample reweighting and precision-oriented threshold selection.
- Computing: Python, C/C++, ROOT/RooFit, NumPy, pandas; TensorFlow/GPU code adaptation and verification; Linux and shared GPU workstation setup.

Research experience

Postdoctoral Researcher | 2021–2026

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Developed a Shell pipeline for beam-test acquisition, storage, quick calibration and initial quality checks, connecting detector readout, tracking and timing calibration. Participated in on-site setup and data taking; assisted subsequent fitting and measurement-effect corrections.
- Connected beam-test measurements and position-dependent response maps to simulated energy deposits, evaluating performance against measured rather than ideal response.
- Developed time-resolved signal analysis and ROOT/C++ correction workflows; checked reconstructed energy before and after correction under the studied conditions.
- Diagnosed incorrect ECAL photon truth associations; revised cluster-energy matching, tested multiple simulated decay channels and improved thread safety through review. The merged change supported subsequent Run 3 analyses and reconstruction-performance checks.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

Measurement-informed detector model

Developed Shell automation linking beam-test acquisition, storage, rapid calibration and quality checks. Participated in setup and data taking, and assisted later analysis. Connected measured response maps with Geant4 simulation; non-uniformity contributed below 1% to the constant term in the first prototype study.

Signal decomposition and correction

Extended simulated scintillation timing and fitted signal-component fractions to correct reconstructed energy. A representative rotation-based PSD study reduced a resolution term by about 2×; material comparisons exposed separation limits. Results apply to the studied configuration.

ECAL diagnosis and integrated software correction

Traced anomalous simulated photon resolution to incorrect truth associations under high occupancy. Revised matching using normalised truth-particle energy deposits within the photon cluster. Checked identities, energies and event histories across several simulated decay channels. Improved thread safety following review; the MR was merged and used in Run 3 analyses and reconstruction-performance checks.

Learning TensorFlow and delivering GPU computation

Learned TensorFlow from inherited code; personally ported amplitude models, added efficiency, resolution and bias effects, and validated calculations against mature C++ code at identical sampling points. Resolved discrepancies by aligning variable types; checked simulated observables against real data with KS tests. Set up a shared GPU workstation in half a day, then achieved a GPU run in another half-day. Contributed to testing a collaborator-developed fitter.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

Learning, delivery & collaboration

- Learning by implementation: extended unfamiliar TensorFlow code with new research models and checked it on generated data; applied CatBoost and Geant4 in other projects.
- Delivery: agreed component interfaces and variable definitions before coding; reassessed priorities and timelines with collaborators as data processing exposed new issues.
- Collaboration: GitLab development and mutual merge-request review; identified a fitter sign error through mathematical checks. Weekly planning and rapid technical discussions; research presentations at GDR-InF 2022 and FCC/DRD France 2025.

Proposed first assignment: agree the decision and success criteria, check data quality, establish a baseline and deliver a focused validation study with findings and next steps.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- Measurement-informed detector model (https://yingrui-hou.github.io/PersonalPage/index.html#work-4)
- Signal decomposition and correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-5)
- ECAL diagnosis and integrated software correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- Learning TensorFlow and delivering GPU computation (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)